

DDoS Incident: Case Study

Case Study Background

This scenario was provided as part of the **Google Cybersecurity Professional Certificate**. It has been condensed here to provide context for my NIST Cybersecurity Framework incident analysis.

SCENARIO

A fictional multimedia company providing web design, graphic design, and social media marketing services to small businesses experienced a distributed denial-of-service (DDoS) attack that disrupted its internal network for approximately two hours.

During the attack, the company's network services stopped responding because of a large volume of incoming ICMP packets. The traffic prevented normal internal network users from accessing network resources.

The incident management team responded by:

- Blocking incoming ICMP packets
- Taking non-critical network services offline
- Restoring critical network services

INVESTIGATION FINDINGS

The cybersecurity team determined that a malicious actor had flooded the network with ICMP ping requests through an improperly configured firewall. The firewall configuration allowed the attacker to overwhelm the company's network with the DDoS attack.

Following the incident, the network security team implemented:

- A firewall rule limiting the rate of incoming ICMP packets
- Source IP address verification to identify spoofed IP addresses
- Network monitoring software to detect abnormal traffic patterns
- An IDS/IPS system to filter suspicious ICMP traffic

ASSESSMENT OBJECTIVE

Using the incident as a case study, I was tasked with applying the NIST Cybersecurity framework to develop a security strategy addressing the incident through five functions:

- • Identify
- • Protect
- • Detect
- • Respond
- • Recover

My completed analysis documents how I applied each function to the incident and considered improvements that could help the organization prevent, detect, respond to, and recover from similar security incidents.